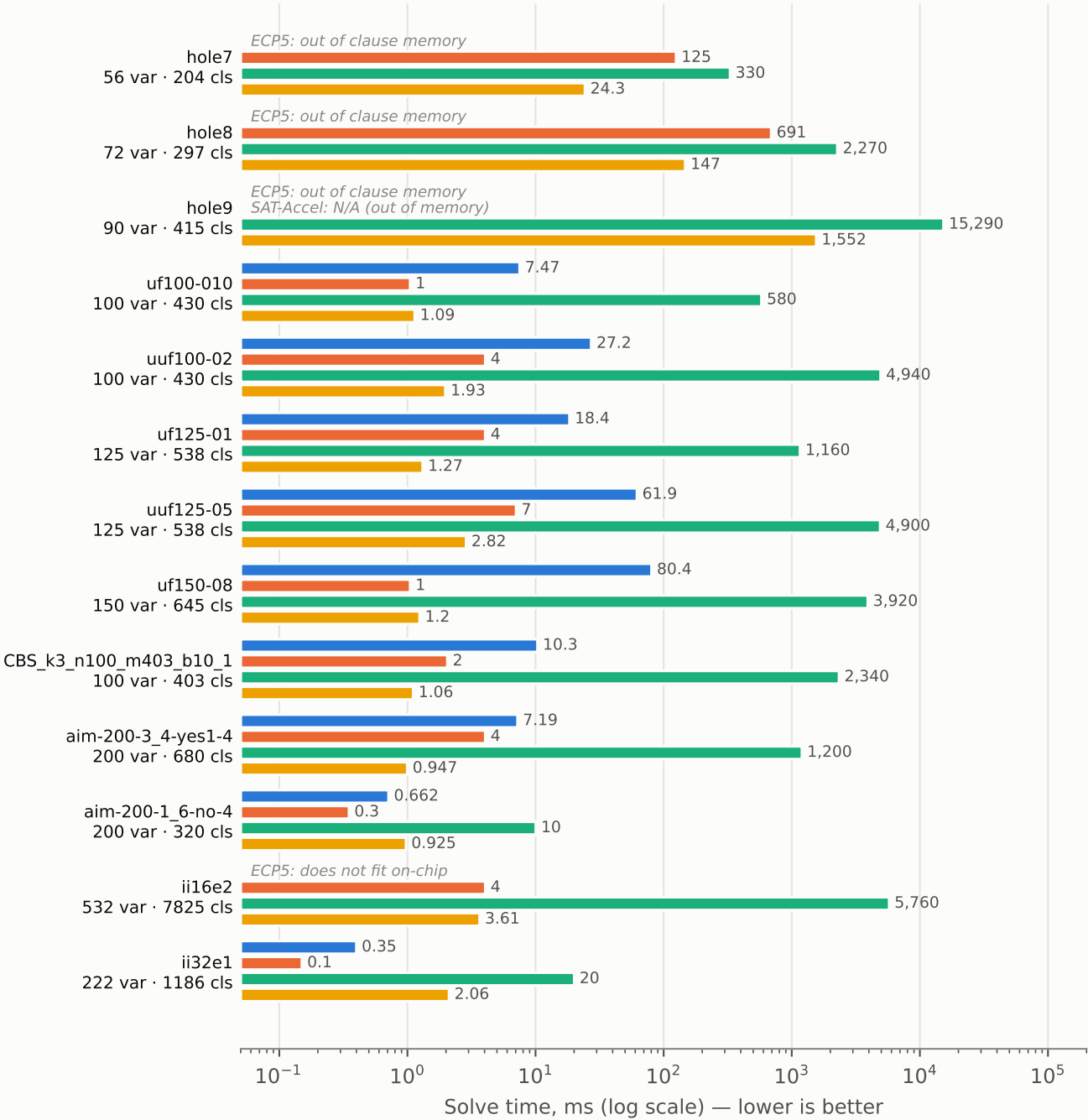


SAT-Accel reimplemented in Amaranth on a Lattice ECP5-85F

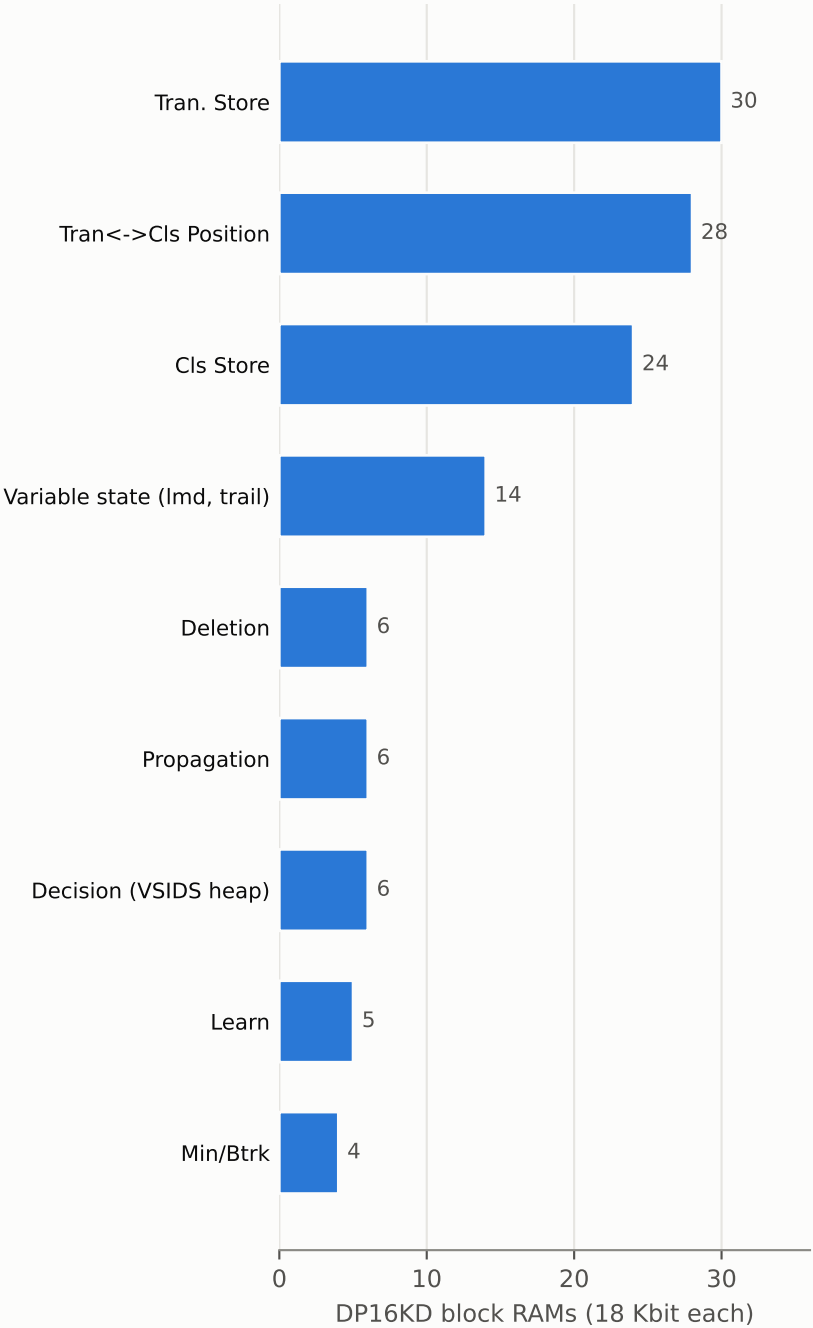
(a) Table 3 — SATLIB instances used by SAT-Hard

This work — ECP5-85F @ 50 MHz SAT-Hard (paper)
SAT-Accel — U55C @ 230 MHz (paper) MiniSat 2.2 — Apple M4 Pro (measured)



(b) Table 2 — block RAM by module

solver 123 / 208 DP16KD · full SoC 147
U55C SAT-Accel: 419 BRAM + 778 URAM



ECP5: measured on a Manhattan Reasoning cloud ECP5-85F (fpga1), on-chip solve cycles at 50 MHz, host load excluded; all 9 solved answers correct, SAT models re-verified. Geomean: 78× faster than SAT-Hard, 6× slower than MiniSat on the M4 Pro. SAT-Accel and SAT-Hard times from Lo et al., FPGA '25, Table 3. MiniSat 2.2 default options, median CPU time of 11 runs (includes parsing; ~1 ms floor). Resources: ECP5 solver 10.2K LUT, 4.0K FF, 3 DSP, Fmax 63.5 MHz; the ECP5 has no URAM, so the clause / transposed stores and position maps (67% of solver BRAM) live in block RAM.